



Attention-guided hierarchical fusion U-Net for uncertainty-driven medical image segmentation

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ABSTRACT

Small inaccuracies in the system components or artificial intelligence (AI) models for medical imaging could have significant consequences leading to life hazards. To mitigate those risks, one must consider the precision of the image analysis outcomes (e.g., image segmentation), along with the confidence in the underlying model predictions. U-shaped architectures, based on the convolutional encoder-decoder, have established themselves as a critical component of many AI-enabled diagnostic imaging systems. However, most of the existing methods focus on producing accurate diagnostic predictions without assessing the uncertainty associated with such predictions or the introduced techniques. Uncertainty maps highlight areas in the predicted segmented results, where the model is uncertain or less confident. This could lead radiologists to pay more attention to ensuring patient safety and pave the way for trustworthy AI applications. In this paper, we therefore propose the Attention-guided Hierarchical Fusion U-Net (named AHF-U-Net) for medical image segmentation. We then introduce the uncertainty-aware version of it called UA-AHF-U-Net which provides the uncertainty map alongside the predicted segmentation map. The network is designed by integrating the Encoder Attention Fusion module (EAF) and the Decoder Attention Fusion module (DAF) on the encoder and decoder sides of the U-Net architecture, respectively. The EAF and DAF modules utilize spatial and channel attention to capture relevant spatial information and indicate which channels are appropriate for a given image. Furthermore, an enhanced skip connection is introduced and named the Hierarchical Attention-Enhanced (HAE) skip connection. We evaluated the efficiency of our model by comparing it with eleven well-established methods for three popular medical image segmentation datasets consisting of coarse-grained images with unclear boundaries. Based on the quantitative and qualitative results, the proposed method ranks first in two datasets and second in a third. The code can be accessed at: <https://github.com/AfsanaAhmedMunia/AHF-Fusion-U-Net>.

1. Introduction

Medical image segmentation (MIS) classifies image pixels semantically [1] according to the local geometric structure, their visual consistency or radiometric properties. In practice, medical image segmentation (MIS) presents challenges for both expert radiologists and computer algorithms. However, it underpins many critical diagnostic radiology examinations, such as quantitative evaluation of organ function, determining the extent and grade of organ lesions, post-treatment

recovery after radiation, hyperthermia, ablation, etc. [2]. Furthermore, the success or failure of MIS has a consequential impact on downstream tasks associated with clinical diagnosis, surgical planning, radiation therapy planning, etc. However, the relationship between MIS and subsequent tasks is as complex and dominant as it is unclear and unexplored [2]. Various medical imaging modalities can be used as input for medical image segmentation, including PET [3], SPECT [4], CT [5], and MRI [6]. Each modality has inherent aleatoric (measurement) errors

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and epistemic (causal) uncertainty that remain unaccounted for by many predictive AI algorithms underpinned by medical imaging.

Initially, traditional methods such as edge detection, thresholding, region-based methods, clustering, etc., were commonly used to segment images [7]. Today, these methods are limited to straightforward segmentation tasks. As opposed to traditional image analysis, deep learning technologies offer significantly improved abilities to handle large and complex data sets [8]. More specifically, Convolutional Neural Networks (CNNs), have remarkably succeeded in medical image segmentation. For example, Fully Convolutional Networks (FCNs) [9], Cascaded FCN [10], U-Net [11], U-Net++ [12], V-Net [13], SegNet [14] have shown excellent performance in diversified medical imagery and applications. CNNs have been extended and improved in several ways. For instance, combining or integrating features from different scales or levels greatly enhances a model's representational power and performance. By using skip connections, it is possible to utilize the features of the encoder during upsampling on the decoder side and combine them accordingly. Hence, almost all encoder-decoder-based segmentation architectures use skip connections to benefit from various image scales, leading to better segmentation accuracy [2].

Although concatenation or addition are popular as basic feature fusion operators, there are situations where attentional feature fusion could lead to improved performance [15]. For example, a wide range of variations in medical image segmentation is possible due to different imaging modes, individual patient traits, and anatomical differences. In this case, the attention mechanism (spatial or channel-wise [16]) can prioritize the most relevant features that may ensure robust segmentation results [17]. The attention mechanism emphasizes relevant information in an image while suppressing irrelevant ones. This is similar to how we often pay more attention to a few words of a sentence or look at specific parts of the image while describing some text or image [18]. Oktay et al. [19] propose Attention Gates (AG) within their "Attention U-Net" architecture. This attention mechanism discards irrelevant regions from an input image, while emphasizing dominant features.

Most deep learning models today achieve higher accuracy for segmenting images yet overlook a significant aspect: providing confidence estimates (certainty vs uncertainty) behind model predictions or, equivalently, accounting for model's uncertainty. Despite the success of deep learning models, it is difficult to answer the question 'How confident is the model about its generated predictions?' Determining how confident these techniques are in their outcomes (predictions) is an important domain of research in Artificial Intelligence (AI), particularly in deep learning and machine learning. Thus, it demands developing novel methodologies to comprehend and measure the certainty (confidence) of various deep learning and machine learning models. Therefore, measuring the uncertainty in the predicted result of image segmentation can produce more detailed interpretable output necessary to achieve user trust, for instance, in the case of ambiguous medical images where fine-drawn structures or regions are indistinguishable. In this situation, radiologists could provide more accurate results if they provide the predicted results with an uncertainty map [20]. Abdar et al. [21] presented a review of uncertainty quantification methods used in various deep learning methods to measure uncertainty in different fields.

In this paper, we integrate spatial and channel-wise attention and Hierarchical Attention-Enhanced (HAE) skip connections with the basic U-Net model. Spatial attention can help identify intricate relationships and structures in medical images. On the other hand, channel-wise attention emphasizes the interchannel relationships and dependencies. Overall, the dual-attention mechanisms and HAE skip connections enhance the model's ability to capture long-range dependencies and a more comprehensive understanding of the entire image, which ensures richer segmentation. Furthermore, We propose an uncertainty-aware version of our proposed AHF-U-Net that is trained with a specially designed loss function and can provide prediction results with additional uncertainty scores. However, it is crucial to understand that

measuring uncertainty during training does not guarantee an increase in accuracy or Intersection over Union (IoU) scores. Instead, it offers a quantification of certainty for each prediction. We specifically chose three datasets: the breast ultrasound image dataset [22] and two sets of skin lesion segmentation datasets (ISIC16 [23] and ISIC17 [24]), which include low-resolution images and unclear edges, reflecting real-world medical conditions.

2. Related work

Taking into account that we have developed a MIS model that incorporates a U-shaped CNN model with attentional feature fusion and also provides an uncertainty-aware version of it, we divide this literature review section into three sub-sections: Medical Image Segmentation (MIS), Attentional Feature Fusion (AFF) and Uncertainty of MIS. Inside the MIS sub-section, we discuss several CNN-based segmentation techniques that have shown promising results in medical image segmentation. We summarize the different types of attention mechanisms and the use of attention mechanisms in medical image segmentation in the following sub-section. Lastly, we review some previously established works that consider uncertainty quantification in MIS.

2.1. Medical Image Segmentation (MIS)

The U-Net architecture [11], which serves as a milestone for medical image segmentation, consists of an encoder-decoder architecture. The nature of this type of architecture is to reduce and expand the spatial dimension of the input image. On the encoder side, spatial dimensions are reduced, and contextual information is captured simultaneously while the decoder expands spatial dimensions. Another unique trait of this type of architecture is skipping connections. This makes it possible to combine low- and high-level features to generate a detailed segmentation map.

Following U-Net's success, several methods were developed that either incorporated U-Net as a backbone architecture or modified it to meet specific requirements. For example, U-Net++ [12] follows the same architecture as U-Net but replaces the basic skip connections with nested and dense skip connections. This architecture is more applicable for reusing the lower-layer features and measuring the semantic gap between the feature maps. However, skip connections in U-Net++ [12] can connect encoders and decoders at the same scale, while a full-scale skip connection is provided in U-Net 3+ [25]. In this type of skip connection, feature maps from the same and smaller scales on the encoder side can be connected to the decoder side. Furthermore, the redesigned U-Net 3 + decoder network [25] is interconnected within decoder subnetworks. These improvements allow U-Net 3+ [25] to incorporate better multiscale features that improve segmentation accuracy. Fig. 1 shows the basic architecture of these three and our proposed architecture. Similarly, CU-Net [26] establishes a connectivity pattern among several stacked U-Nets so that information can flow more efficiently between multiple U-Nets. "U-Net and its variants for MIS" is reviewed by Siddique et al. [27] and Ummadi [28].

Additionally, few architectures focus on 3D imaging, e.g., H-DenseU-Net [29] consists of a 2-D DenseU-Net and its 3-D counterpart for hierarchically aggregating volumetric contexts in the spirit of the auto context algorithm for liver segmentation, 3D U-Net [30] replaces all 2D operations of U-Net with their 3D counterpart to segment the 3D kidney of *Xenopus*, V-Net [13] is another popular network for 3D image segmentation which can be trained end-to-end on MRI volumes for depicting prostate and also handles class imbalance problem with the integration of dice coefficient-based loss function.

2.2. Attentional feature fusion

In general, medical images contain complex structures and fine-drawn irregularities. Therefore, to achieve more precise and meaningful

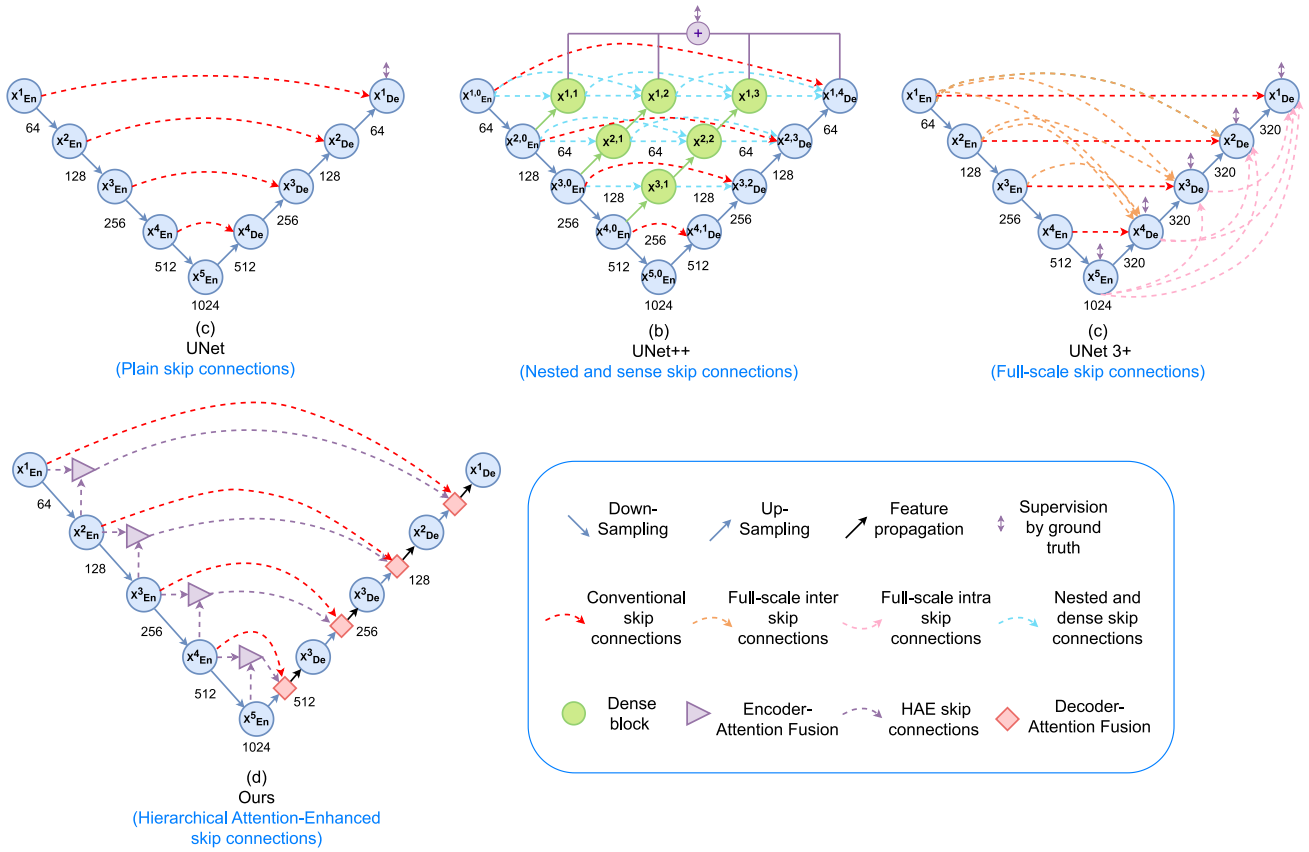


Fig. 1. U-Net models proposed in the literature: (a) Original U-Net [11], (b) U-Net++ [12], (c) U-Net 3+ [25], and (d) Ours.

segmentation, considering both local and global contexts could be beneficial. Here, the attention mechanism balances local details with global context. It showed a significant improvement in the precision and IOU score of the predicted segmented results, e.g., [17,19,31,32]. However, their significance can fluctuate depending on the application for which they are designed and the challenges they must overcome. For example, the local context can preserve adjacent tissues or lesions. It can also reduce the noise reduction of images or help to discover tiny lesions or tumors [33]. In contrast, the global context helps alleviate the size and shape restrictions of organs or lesions and can describe anatomical relationships between organs [34]. This also considers the variability of medical images. Dai et al. [15] accumulated local and global contexts for image segmentation in their proposed 'Attentional feature fusion' mechanism. They incorporate it into the architecture of ResNet [35], InceptionNet, and FPN [36]. Channel-U-Net [32] integrates channel-wise attention within each encoder and decoder of the U-Net. Create a relational map between the pixels extracted from channel-wise convolution and the information extracted from the feature maps, achieving multiscale information fusion. Similarly, SCAU-Net [31] integrates channel-specific spatial attention with U-Net to amplify local characteristics while eliminating irrelevant ones. TransAttUnet [17] is a transformer-based segmentation network consisting of Transformer Self-Attention (TSA) and Global Spatial Attention (GSA), which provides multilevel guided attention. In this research, we have meticulously analyzed the methods discussed so far and employ a combination of spatial and channel-wise attentional feature fusion in multiple layers of our proposed architecture.

2.3. Uncertainty in MIS

Undeniably, deep learning-based automated segmentation methods provide sophisticated results that ease radiologists' and clinicians'

work. At the same time, it is essential to understand that high accuracy does not imply high confidence or certainty about the predicted results. In addition, uncertainty is crucial when even a slight prediction error could be hazardous. Therefore, an ideal model should deliver accurate predictions as well as signal the exigency for augmented scrutiny where it is uncertain. This is necessary to prevent misdiagnosis [37]. Even uncertainty estimation has also shown a noticeable impact on image classification [38–40], etc. Abdar et al. [41] argued that understanding and quantifying the inherent uncertainty in MIS makes it possible to ensure reliable prediction and hence achieve the trust of clinicians.

Kohl et al. [42] proposed an approach for segmenting ambiguous images by integrating the probabilistic approach with the conventional U-Net model. In addition to providing a singular, deterministic segmentation prediction, this model provides a distribution of potential segmentation maps for a given input image. Gal and Ghahramani [43], Kendall and Gal [44], and Postels et al. [38] also outlined probabilistic approaches to deal with epistemic uncertainty about the expected results. The ensemble technique has been used extensively for uncertainty estimation. Here, the prediction depends on the average of multiple results, which can be related to multiple annotations of medical images. For instance, for the proper annotation of medical images, especially ambiguous images, multiple annotations of a single image are done by multiple experts, and then analyzing the agreement and disagreement, one can understand which areas are highly uncertain and which are not. Wenzel et al. [45] proposed an ensemble-based model for uncertainty estimation by creating various hyperparameter settings and assessing their variations in prediction. Similarly, the multidecoder U-Net [46] compares predictions derived from multiple decoders. Here, the authors utilized dice loss as a similarity metric and estimated high and low uncertainty in the final output. Using the ensemble of classifiers [47], Lu et al. [48] assess the uncertainty for Semi-supervised Medical Image Segmentation. They have developed an

uncertainty-informed prototype consistency learning framework called UPCoL, which considers voxel-level consistency in latent features and decision space. The uncertainty quantification for each voxel is represented through an entropy-based measure by averaging four prediction results from four classifiers.

Monte Carlo Dropout approach [43] can also be used to estimate the uncertainty of a deep learning model. To improve needle placement accuracy in Point-of-Care Ultrasound (POCUS), Arun et al. [49] propose a technique called Temporal Monte Carlo dropout. The method reduces computational complexity by 8x without compromising segmentation performance, achieving an 84% similarity score to the Monte Carlo Dropout. To accurately assess MRI-iUS (Intra-operative ultrasound) registration errors for brain tumor surgery, Salari et al. [50] propose a 3D neural network called FocalErrorNet that is also capable of estimating uncertainty. The architecture of this focal modulation network includes two MC dropout layers to the multilayer perceptron (MLP) to facilitate uncertainty quantification of the results. At test time, 200 regression samples from MC dropouts were randomly sampled for each MRI-iUS patch pair. The mean and standard deviation of these samples are used as the final patch registration error and the uncertainty metric, respectively.

Sensoy et al. [51] propose an evidential deep learning technique to measure classification uncertainty employing a Dirichlet distribution on class probabilities. By using this method, Zou et al. [52] developed a reliable medical image segmentation framework, EvidenceCap, which can be easily integrated with any backbone network. In their research, they integrated EvidenceCap with U-Net, AU-Net, and V-Net and showed noticeable performance improvements on three popular medical image datasets.

After a thorough analysis of the aforementioned uncertainty quantification techniques, we have implemented MC Dropout, Deep Ensemble, and EvidenceCap methods for our selected datasets. We assessed their performance both quantitatively, by measuring the Intersection over Union (IoU) and Dice coefficients for certain pixels versus all pixels, and qualitatively, through observation of UQ maps. We also introduced several modifications to the original EvidenceCap method. Ultimately, we found that our modified EvidenceCap achieved more sophisticated quantitative and qualitative results compared to the other two methods. Furthermore, it is a computationally efficient solution that closely aligns with the requirements of our segmentation model. Detailed results and analyses are provided in Section 5.

3. Methodology

In this section, we show how we designed our AHF-U-Net architecture. First, we discuss the overall architecture and the main components of the model. Next, we provide the structure and goal of Encoder Attention Fusion (EAF) and Decoder Attention Fusion (DAF). Following that, the uncertainty-aware version of AHF-U-Net is explained. We used different loss functions for AHF-U-Net and uncertainty-aware AHF-U-Net, as discussed at the end of this section.

3.1. Overview of AHF-U-Net

The proposed AHF-U-Net is an encoder–decoder-based architecture that employs attention mechanisms on the U-shaped network's encoder and decoder sides. We also provide the uncertainty-aware version of AHF-U-Net to measure the confidence of predicted results. The main difference between them is that the uncertainty-aware version provides automatic segmentation of medical images according to their pixel-wise semantic label and offers an uncertainty measure of prediction. Therefore, if the model receives an input image of $X \in \mathbb{R}^{C \times H \times W}$, it should produce two outputs: a predicted segmented map and its corresponding uncertainty map, both with dimensions $H \times W$. Here, C represents the color channel in the input layers and the feature channel in the subsequent layers, while $H \times W$ represents the spatial dimension

of the image. The general structure of the learning process is shown in Fig. 2. The Encoder Attention Fusion (EAF) module, the Decoder Attention Fusion module (DAF), and Hierarchical Attention Enhanced (HAE) skip connections are the main contributions of this network. Each EAF module takes input from two subsequent encoder layers. On the other hand, each DAF module takes three inputs: one from the EAF module output and the other two from its corresponding encoder and decoder layer. We employ two attention mechanisms: the spatial and channel attention mechanisms in both the EAF and DAF modules. This enables the model to capture relevant spatial details and to emphasize which channels are most relevant to the input image. The spatial attention mechanism focuses on the most prominent features in spatial dimensions and captures the global context of the image (broader contextual information and patterns) [16]. Meanwhile, the channel attention mechanism focuses on the most significant features between channels and determines the relationships between different channel characteristics [16]. Hence, by leveraging both attention mechanisms, the model can produce a more comprehensive feature representation for understanding complex structures in medical images. Our proposed HAE skip connection integrates attention-enhanced features from two subsequent encoder layers that ensure encapsulation of both local details and the global context of the image. On the other hand, simple skip connections transfer fine-grained spatial information directly from the encoder to the decoder, which helps the model generate more precise segmentations [11]. The benefit of using HAE and simple skip connections is that the decoder can have contextual information and detailed boundaries of intricate structures in medical images. As a result, AHF-U-Net can produce an accurate and smooth segmentation map.

3.2. Encoder Attention Fusion (EAF)

The EAF module takes two consecutive encoder outputs as input. In this way, four EAF modules have been placed for five encoder layers in the network. For instance, EAF(1,2) integrates input from encoder block-1 ($F_{en}^1 \in \mathbb{R}^{C_1 \times H_1 \times W_1}$) and encoder block-2 ($F_{en}^2 \in \mathbb{R}^{C_2 \times H_2 \times W_2}$) that provides an attention-wise fused feature ($F_{hae}^{1,2}$) as output. This integrated output is denoted as HAE(1,2) in Fig. 2. Similarly, EAF(2,3), EAF(3,4) and EAF(4,5) follow the same principles and produce outputs represented by HAE(2,3), HAE(3,4) and HAE(4,5).

Fig. 3 represents a detailed overview of EAF(1,2). Inside this module, a global spatial pooling, $g_s(F) = \frac{1}{H \times W} \sum_{i=1}^H \sum_{j=1}^W (F)_{[i,j]}$, and a global channel-wise pooling, $g_c(F) = \frac{1}{C} \sum_{c=1}^C (F)_{[c,:]}$ are performed on each feature map. To be more specific, these pooling operations are applied to $F_{en}^1 \in \mathbb{R}^{C_1 \times H_1 \times W_1}$ to generate spatial attention weights (w_s) and channel attention weights (w_c). These weights (w_s, w_c) are then multiplied together (w_{sc}) to capture both the global context and the inter-channel relationships of the features. The same approach is carried out for $F_{en}^2 \in \mathbb{R}^{C_2 \times H_2 \times W_2}$. Moreover, this step can increase the discriminative power of the model by paying more attention to the salient features while giving less priority to irrelevant features.

We have performed element-wise addition among these multiplied features to achieve multiscale feature fusion, F_{fused} . To effectively perform element-wise addition, feature dimensions must be compatible, hence we apply transpose convolution ($TConv$) to the features received from encoder block-2. According to the structure of U-Net, each encoder layer stores features at a different scale. To be more specific, encoder block-1 captures fine-grained, low-level details. In contrast, encoder block-2 provides relatively high-level features, followed by encoder block-5, which offers more abstract, high-level features (broader structures and patterns) [11]. Therefore, combining two consecutive layers can ensure a more comprehensive representation of features. This fused feature is then fed into a convolution operation followed by $\text{relu}(\delta)$ activation to obtain a feature map that preserves the spatial and channel attention weights. The convolution operation maintains the same channel and spatial dimensions for input and output. The

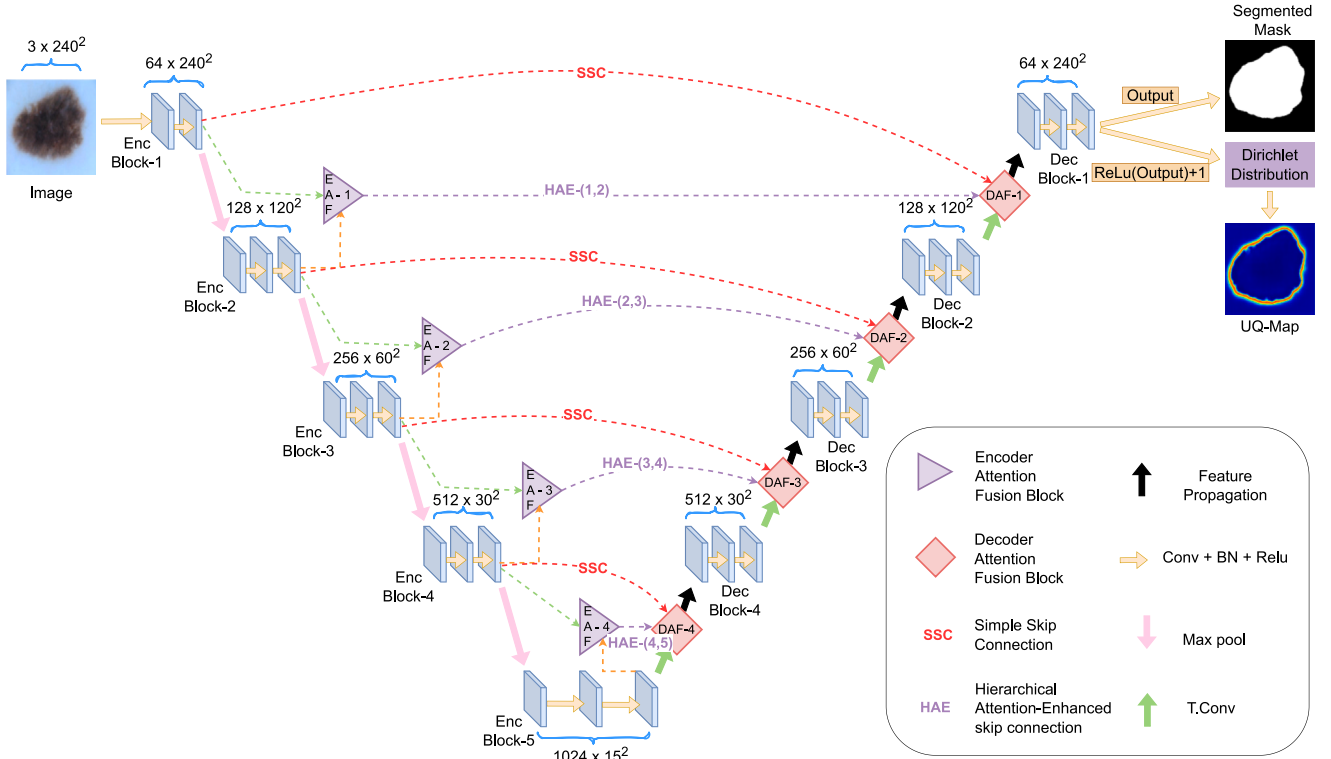


Fig. 2. Detailed view of the proposed model.

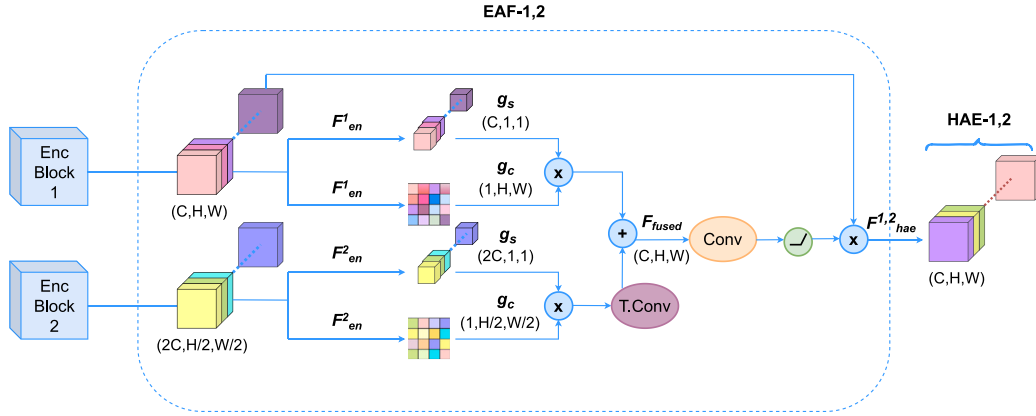


Fig. 3. Encoder attention fusion block.

advantage of keeping the channel number is increasing network depth or refining features without Additional Parameters [53]. Furthermore, the preservation of spatial dimension makes it possible to capture local spatial patterns, which are essential for segmenting intricate medical organs [54]. In the last step, we multiply the generated attention weights with the encoder block-1 feature (F_{en}^1) map so that the EAF block can deliver hierarchical attention-enhanced features to its corresponding decoder block. This process can be mathematically described as follows (Here, $c_2 = 2C_1$, $H_2 = \frac{H_1}{2}$ and $W_2 = \frac{W_1}{2}$):

$$w_{s \times c}^1 = g_s(F_{en}^1) \times g_c(F_{en}^1) \quad (1)$$

$$w_{s \times c}^2 = g_s(F_{en}^2) \times g_c(F_{en}^2) \quad (2)$$

$$F_{fused} = w_{s \times c}^1 + TConv(w_{s \times c}^2) \quad (3)$$

$$F_{hae}^{1,2} = F_{en}^1 \times \delta(Con(F_{fused})) \quad (4)$$

3.3. Decoder Attention Fusion (DAF)

We include a DAF module before every decoder block in the network, so for four decoders, we position four DAF modules. This way, each decoder layer can get more comprehensive and relevant information about the image, leading to an enhanced segmentation map. Here, we propose two different fusion techniques to explore and compare different strategies for enhancing decoder attention fusion in the AHF-U-Net model: updated-AFF and CLSCA, both performing remarkably well.

We developed the Updated AFF (UAFF) architecture to leverage the established benefits of attentional feature fusion (AFF) proposed by Dai et al. [15]. The AFF module can fuse features of varying semantics and scales, making it suitable for common feature fusion scenarios such as long and short skip connections. By adapting proven frameworks, we ensure a robust foundation for feature fusion through attention mechanisms.

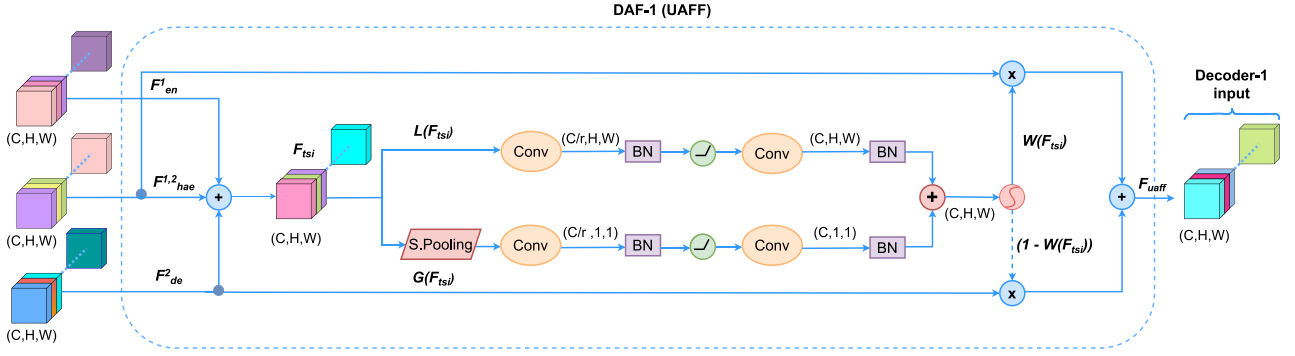


Fig. 4. Decoder attention fusion block based on UAFF fusion approach.

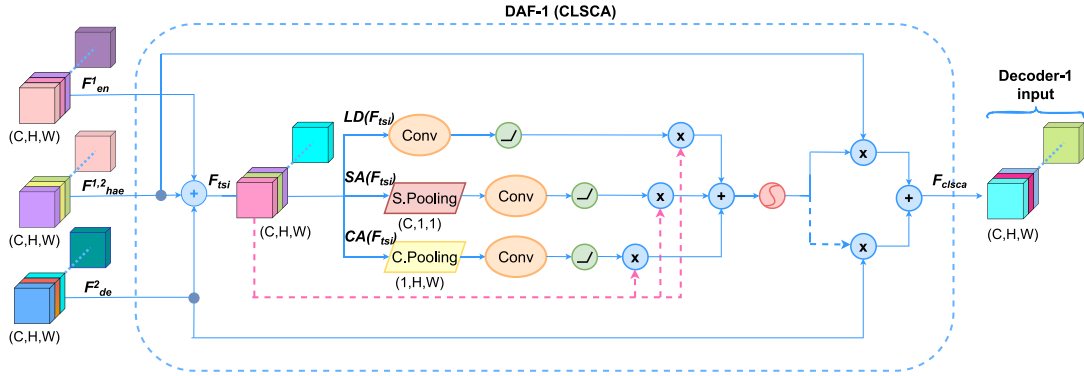


Fig. 5. Decoder Attention Fusion Block based on CLSCA fusion approach.

In contrast, the CLSCA module stands for Capturing Local patterns with Spatial and Channel Attention. It is proposed to investigate the benefits of incorporating both spatial and channel-wise attention across different levels of the network. By developing CLSCA, we aim to demonstrate the added value of incorporating dual attention mechanisms that are specifically crucial for identifying complicated patterns in medical images. Both modules take the same inputs and follow the same initial fusion technique.

For instance, Figs. 4 and 5 demonstrate that they take three inputs and fuse them, such as:

- Features from the corresponding encoder layer, F_{en}
- Hierarchical Attention-Enhanced features from the corresponding EAF module, F_{hae}
- Upsampled decoder features from the previous decoder layer, F_{de}

$$F_{tsi} = F_{en} + F_{hae} + F_{de}$$

The combination of these three features produces a new part that captures local details (from encoder features), global context (from attention-enhanced elements), and spatial information (from upsampled decoder features). This extensive feature is denoted as the Triple-Stream Integrated (TSI) feature, which is designated as F_{tsi} in Figs. 4 and 5. However, the rest of the steps of UAFF and CLSCA are different and will be discussed in the following two subsections.

3.3.1. Overview of UAFF

Fig. 4 shows a complete overview of the UAFF-based DAF module. For simplicity, let us consider the DAF-1 module within the AHF-U-Net network. The UAFF takes three feature maps as input (discussed in Section 3.3) and performs element-wise addition among them, unlike AFF, which takes two feature maps as input. After this initial integration, the TSI feature is generated and passed through two branches:

one for capturing local channel context $L(F_{tsi})$ and the other for global channel context $G(F_{tsi})$.

$$L(F_{tsi}) = \beta \left(PCov_i \left(\delta \left(\beta \left(PCov_r \left(F_{tsi} \right) \right) \right) \right) \right) \quad (5)$$

$$G(F_{tsi}) = \beta \left(PCov_i \left(\delta \left(\beta \left(PCov_r \left(g_s \left(F_{tsi} \right) \right) \right) \right) \right) \right) \quad (6)$$

where β represents Batch Normalization (BN), δ represents ReLu and $g_s(F_{tsi})$ represents global spatial average pooling on feature F_{tsi} . All the convolutions used in these two branches are point-wise convolutions ($PCov$: kernel size 1×1) where $PCov_r$ reduces the channels from C to C/r and $PCov_i$ increases the channel from C/r to C (r = reduction ratio). The attention weight W is obtained by performing a broadcasting addition ($\oplus$) of $L(F_{tsi})$ and $G(F_{tsi})$ and passing the result through Sigmoid activation (σ) to scale it between 0 and 1.

$$W(F_{tsi}) = \sigma \left(L(F_{tsi}) \oplus G(F_{tsi}) \right) \quad (7)$$

A feature recalibration is done by multiplying $W(F_{tsi})$ with the HAE feature (F_{hae}) and $(1 - W(F_{tsi}))$ with the upsampled decoder features (F_{de}). Combining these features completes the final fusion of UAFF, and a final refined feature is prepared to feed the corresponding decoder.

$$F_{uaff} = W(F_{tsi}) \times F_{hae} \oplus (1 - W(F_{tsi})) \times F_{de} \quad (8)$$

3.3.2. Overview of CLSCA

Fig. 5 shows a complete overview of the CLSCA-based DAF module. For simplicity, consider the DAF-1 module, which is placed between Decoder blocks 1 and 2 in the original network Fig. 2. In CLSCA, the initial integration technique and input features are the same as in UAFF. However, here, we use three branches to focus on features in different

aspects instead of the two used in the UAFF. The purpose of these branches is discussed below:

$$LD(F_{tsi}) = \delta(\text{Conv}(F_{tsi})) \times F_{tsi} \quad (9)$$

The branch ‘LD’ stands for ‘Local Details’, which captures immediate local details of the TSI feature.

$$SA(F_{tsi}) = \delta(\text{Conv}(g_s(F_{tsi}))) \times F_{tsi} \quad (10)$$

The branch ‘SA’ signifies Spatial Average pooling, which applies global average pooling along the spatial dimension (g_s) of TSI features, and the result is fed into a convolution and ReLU activation to generate a channel-wise attention map. After that, an element-wise multiplication is done between this channel-wise attention map and the original TSI features, which provides channel-weighted features.

$$CA(F_{tsi}) = \delta(\text{Conv}(g_c(F_{tsi}))) \times F_{tsi} \quad (11)$$

The branch ‘CA’ denotes Channel Average pooling, which applies global average pooling along the channel dimension (g_s), which generates spatial-wise attention features. Combining these enhanced features offers an attention-enhanced feature map that captures both the local details and global information (spatially and channel-wise).

$$W(F_{tsi}) = \sigma(LD(F_{tsi}) + SA(F_{tsi}) + CA(F_{tsi})) \quad (12)$$

$$F_{clsca} = W(F_{tsi}) \times F_{hae} + (1 - W(F_{tsi})) \times F_{de} \quad (13)$$

3.4. Uncertainty-aware AHF-U-Net (UA-AHF-U-Net)

The UA-AHF-U-Net is an uncertainty-sensitive version of AHF-U-Net, measuring the uncertainty in the predicted segmentation mask. The idea we use to measure segmentation uncertainty is similar to EvidenceCap [52], which can easily integrate with any backbone network. EvidenceCap uses the Dempster–Shafer Theory of Evidence (DST) [55] to evaluate uncertainty. DST provides predicted probabilities and ‘belief masses’ for subsets of a ‘frame of discernment.’ The term ‘frame of discernment’ refers to all possible states or outcomes. Assigning a high belief mass to a subset, a low belief mass to others, or even the same belief mass to all the subsets is possible. When the same belief mass is assigned to all subsets, it could be interpreted as “I’m unsure about the prediction”. This belief mass is presented formally through Subjective Logic (SL) [56]. Using the Dirichlet distribution, SL provides a way to quantify belief mass and uncertainty based on evidence. Dirichlet distributions generalize Beta distributions, capable of capturing uncertainty in multinomial distributions [57]. For example, consider that SL provides a belief mass b_c to each frame of mutually exclusive outcome C (C is similar to pixel labels or class labels in segmentation problem, $c = 1, 2, \dots, C$) and an overall uncertainty mass u ; here all values are positive and their sum is equal to one.

$$u + \sum_{c=1}^C b_c = 1, \text{ where } u \geq 0, \text{ and } b_c \geq 0 \forall c = 1, 2, \dots, C \quad (14)$$

For a particular class c , its associated belief mass b_c is determined from its evidence e_c and the total evidence S , where $S = \sum_{i=1}^C (e_i + 1)$. Similarly, the general uncertainty of a prediction can be calculated by dividing the total number of classes by the total evidence.

$$b_c = \frac{e_c}{S} \text{ and } u = \frac{C}{S} \quad (15)$$

Here, we modify Eqs. (14) and (15) specifically for image segmentation, where each pixel (i, j) of an image (X) can be classified into a frame of C mutually exclusive classes.

$$u^{(i,j)} + \sum_{c=1}^C b_c^{(i,j)} = 1 \quad (16)$$

$$b_c^{(i,j)} = \frac{e_c^{(i,j)}}{S} \text{ and } u^{(i,j)} = \frac{C}{S}, \quad (17)$$

where $u^{(i,j)} \geq 0$ and $b_c^{(i,j)} \geq 0$ for $c = 1, \dots, C$. In the context of image segmentation, $e_c^{(i,j)}$ represents the evidence derived for the c th class at pixel (i, j) , and $b_c^{(i,j)}$ represents the belief mass for that class at pixel (i, j) . The uncertainty mass $u^{(i,j)}$ quantifies the uncertainty of the classification at that pixel, providing a measure of confidence in the predicted class label and S is the total evidence calculated with $S = \sum_{c=1}^C (e_c^{(i,j)} + 1)$.

3.5. Loss function

We use two different loss functions for the end-to-end training of our proposed models. First, we train AHF-U-NET (UAFF) and AHF-U-NET (CLSCA) using the Binary Cross Entropy with Logits loss function (BCE-withLogits). This loss function combines Sigmoid activation and Binary Cross Entropy loss function; hence, the input should be the model’s raw output. Eq. (18) shows the formula for calculating BCEwithLogits loss for the segmentation problem:

$$L_{BC} = -\frac{1}{H \times W} \sum_{i=1}^H \sum_{j=1}^W [Y_{i,j} \log(\sigma(\hat{Y}_{i,j})) + (1 - Y_{i,j}) \log(1 - \sigma(\hat{Y}_{i,j}))], \quad (18)$$

where H and W represent the image’s height and width, $Y_{i,j}$ denotes the ground truth label at the (i, j) -th pixel, $\hat{Y}_{i,j}$ denotes the predicted logit at the (i, j) -th pixel and σ is the sigmoid activation function.

EvidenceCap proposed a loss function to calibrate uncertainty in 3D medical images. Following that, we modified it for our 2D image segmentation.

$$L_U = -\beta_t \sum_{i,j \in \{\hat{y}_{ij}=y_{ij}\}} p_{ij} \log(1 - u_{ij}) - (1 - \beta_t) \sum_{i,j \in \{\hat{y}_{ij} \neq y_{ij}\}} (1 - p_{ij}) \log(u_{ij}) \quad (19)$$

where, $\hat{y}_{ij}$ and y_{ij} is the predicted label and true label at the (i, j) -th pixel, respectively; p_{ij} represents the confidence of the prediction at the (i, j) -th pixel, while u_{ij} is the uncertainty for the same pixel. β_t works as an annealing factor where $\beta_t = \beta_0 e^{-(\ln \beta_0 / T)^t}$ and T and t are the total number of epochs and the current epoch, respectively. The purpose of using an annealing factor here is to dynamically adjust the optimization process, which leads to better model generalization and convergence [58].

Finally, to make an uncertainty-aware segmentation model, we combine the uncertainty loss (L_U) function and segmentation loss function (L_{BC}) and train the model with this combined loss function (L_C). So, the two uncertainty-aware versions of AHF-U-Net, UA-AHF-U-NET (UAFF) and UA-AHF-U-NET (CLSCA), are trained with the following loss function:

$$L_C = L_{BC} + L_U \quad (20)$$

4. Experiment

4.1. Experimental dataset

To evaluate the effectiveness of our proposed AHF-U-Net in conditions of low resolution and unclear edge definition, we performed experiments on breast tumor segmentation and skin lesion segmentation. For breast tumor segmentation, we use the breast ultrasound image dataset [22]. For skin lesion segmentation, we use the dataset proposed by the International Skin Imaging Collaboration (ISIC) in 2016 [23] and 2017 [24] (see Fig. 7). Although the core task of both challenges includes automated skin lesion segmentation from dermoscopic images, ISIC-2017 includes a larger and more structured

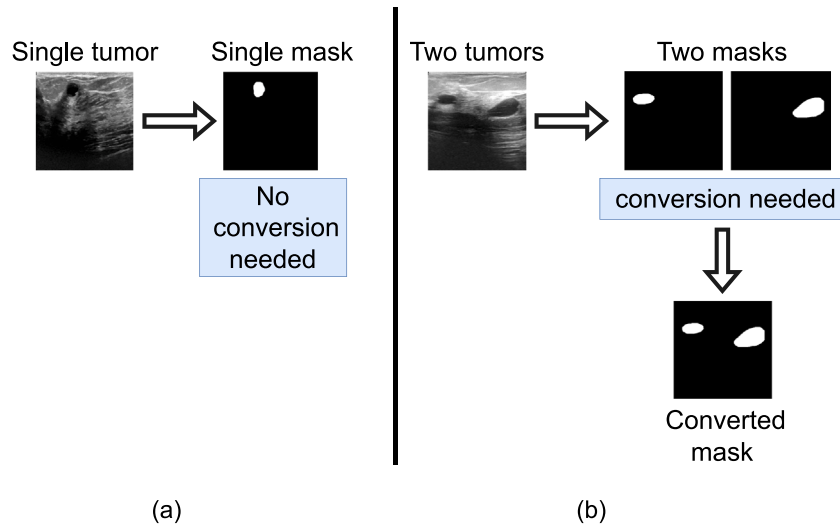


Fig. 6. Mask conversion: (a) The image contains a single tumor, and the dataset provided a single mask for that, so no conversion is needed. (b) The image contains two tumors, and the dataset provided two individual masks for two tumors, so conversion is needed to combine them into one.

dataset with real-time feedback mechanisms. Hence, we utilize both to enhance the robustness and generalizability of our model through more comprehensive training and testing.

(1) **Breast Ultrasound Dataset (BUS):** The data presented in the breast ultrasound image dataset is collected from 600 female patients aged 25 to 75. Initially, 1100 images were gathered, but after preprocessing and removing duplicate images, 780 images were finalized for the BUS dataset. They have categorized all images into three classes: normal, benign, and malignant, and also provide ground truth masks of each image. The number of ground truth masks for each image depends on the number of tumors. For example, if the image contains one tumor, one mask is provided, and if the image has three tumors, three masks are provided. We performed the following preprocessing steps to train our network on this dataset: The images marked as ‘normal’ do not have tumors, hence nothing to segment, as this research focuses mainly on image segmentation, we eliminated all these images and masks. Since the images come in different shapes, we had to resize them all to 240×240 . We need only one output mask corresponding to a single input image to ensure a straightforward training process. Therefore, we combined multiple covers into one following the same approach as shown in the research proposed by Munia et al. [59]. Fig. 6 shows a complete process overview.

(2) **ISIC-2016:** The International Skin Imaging Collaboration (ISIC) presents the challenge of contributing to the analysis of skin cancer (Melanoma) from dermoscopic images. Dermoscopy is a noninvasive imaging technique beneficial for early detection of skin cancer [60]. This challenge was first announced in 2016 and divided into three subchallenges: segmentation of the melanoma lesion, four types of characteristic detection, and three types of disease classification [23]. ISIC-2016 provides 900 train images and 379 test images and supplies the corresponding ground-truth mask generated by dermoscopic experts.

(3) **ISIC-2017:** The data found in ISIC-2017 are an expanded version of ISIC-2016. This is the second challenge of ISIC [24] for detecting melanoma from dermoscopic images, which also divides the challenge into three subchallenges like ISIC-2016. Here, the amount of data is significantly higher than in the previous challenge. It includes 2750 dermoscopic images: 2000 for training, 150 for validation, and 600 for testing. All images come with their corresponding binary masks annotated by dermatological experts, where the pixel value 255 represents the injury’s area and the pixel value 0 represents the background.

4.2. Evaluation metrics

The two most common metrics used for the image segmentation algorithm are the mean Intersection over Union (IoU) and the mean DICE coefficient (DICE) [61]. We use these two metrics as our key evaluation metrics to evaluate the quantitative performance of our proposed model. Although IoU and DICE both measure the overlap between the predicted segmented mask and the ground truth mask, the DICE score is more effective in the case where class imbalance may affect the output [62,63]. The images we use to train and test the proposed model are mostly covered with background pixels. Therefore, the use of a DICE score is appropriate. In addition, we measure pixel-wise accuracy, sensitivity, and specificity to evaluate our model.

4.3. Variations of the proposed model

We have compared four versions of the proposed AHF-U-Net with other state-of-the-art models. According to the fusion strategy of the decoder side, we have created two variations: one is based on UAFF, and the other is based on CLSCA. To measure the uncertainty of the predicted result and provide a corresponding uncertainty map with the predicted segmented map, we have trained the two versions of AHF-U-Net (UAFF and CLSCA) with an uncertainty-aware loss function and got two Uncertainty-Aware versions. Thus, we have the following four versions:

- AHF-U-Net (UAFF)
- AHF-U-Net (CLSCA)
- UA-AHF-U-NET (UAFF)
- UA-AHF-U-NET (CLSCA)

4.4. Comparative models

The architecture of AHF-U-Net features a U-shaped architecture that integrates attention-guided and multiscale/hierarchical fusion approaches on both the encoder and decoder sides. To ensure a fair comparison, we carefully considered these features when selecting other models for evaluation. We examined eleven state-of-the-art models, assessing both their quantitative and qualitative performance. These models are categorized into three major groups and distinguished by different colors in all evaluation tables.

As U-Net [11] is foundational in medical image segmentation and our model employs similar downsampling and upsampling processes,

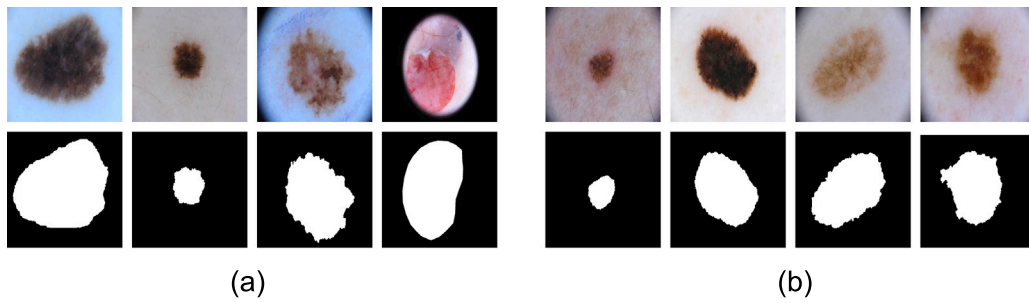


Fig. 7. Dermoscopic images of skin lesions and corresponding masks (a) The images and masks from ISIC-2016 [23]. (b) Images and masks from ISIC-2017 [24].

we included it in our evaluation. SegNet [14], another typical architecture for image segmentation like U-Net (Ronneberger et al. 2015), was also selected for comparison. Models such as U-Net++ [12], UNext [64], CUNet [26], and CANet [65] were chosen due to their use of multiscale skip connections and varied connectivity patterns between encoders and decoders. Additionally, we selected models like AFF [15] and iAFF [15], which emphasize attention guidance, as well as TransUnet [66] and SwinUnet [67], which are based on transformer technology.

4.5. Implementation details

We have implemented the code in PyTorch [68] and run it in Google Colab-Pro [69], utilizing NVIDIA T4 GPU in our experiments. All the datasets we examined for this research are split into training, validation, and test sets. All images are resized to 240×240 pixels to feed our proposed model and other models used for evaluation to make a fair comparison.

We utilized various data augmentation techniques in the training dataset to improve the model's robustness and reduce overfitting. For instance, for random rotations, translations, scaling, and shearing we apply RandomAffine, for randomly zooming into images RandomZoom is applied, and to introduce variation in image orientation we utilize RandomHorizontalFlip and RandomVerticalFlip. To maintain consistency in the input images, we use ResizeTransform in both the training and validation sets.

To find which optimizer and set of hyperparameters best fit our model, we employ a hyperparameter optimization framework called Optuna. It automates the process of searching for the most suitable hyperparameters by analyzing predefined ranges of values. In order to find optimal solutions, it uses a Bayesian optimization technique to narrow down the search space over successive trials [70].

The core of Optuna's optimization process is the objective function, which can be designed for maximization or minimization, depending on the model's requirements. For our segmentation model, we have configured the objective function to maximize the value of the IoU metric. To improve model performance, Optuna executes this function in each trial with a new set of hyperparameters, which are adjusted based on the outcomes from previous trials. We provided three options for the optimizer: Adam, RMSprop, and SGD, and tested extensive ranges for hyperparameters such as learning rate (from 2×10^{-5} to 1×10^{-2}), weight decay (from 2×10^{-5} to 1×10^{-2}), and dropout rate (from 0.0 to 0.5). Recognizing that the learning rate can significantly affect the model's ability to fine-tune its parameters, we set the learning rate scheduler parameters to range from 1 to 10 and established a gamma value range from 0.1 to 1.0.

While Optuna facilitates the search for optimal hyperparameters, it is important to note that the process can be time-consuming due to the number of trials and the comprehensive testing of each parameter set. Nevertheless, this thorough search process is crucial, as it significantly contributes to the accuracy and robustness of the final model. In this study we created an Optuna study with 60 trials. For all three datasets,

Optuna finds Adam to be the most efficient optimizer, with the best initial learning rate of 2.0×10^{-4} for the BUSI dataset and 1.0×10^{-5} for the ISIC-16 and ISIC-17 datasets.

We train the model for 100 epochs and set the batch size to 8. During training, we monitor the model's performance on the validation dataset after every epoch. At the beginning, we set the maximum IoU validation to 0 and continuously updated it with the highest value that was provided by the model at the latest time. An early stopping mechanism can terminate training if this value has not been updated for 20 epochs. The final model saves the weights at the point where the best IoU validation score is achieved.

5. Results and discussion

5.1. Evaluation on breast ultrasound data

Table 1 shows the quantitative results for five evaluation metrics, where red text represents the highest score and green represents the second-highest score. Both versions of AHF-U-Net outperform other state-of-the-art methods concerning the main evaluation metrics (IoU and DICE), among which we benchmark AHF-U-Net (CLSCA) for the BUS dataset. Among the eleven methods we evaluated for this experiment, SwinUnet [67] achieved the highest overall performance across most metrics. However, AHF-UNET (CLSCA) provides approximately 1.2% better IoU (70.12%) and 1.9% better DICE score (82.43%) compared to SwinUnet. In terms of the other three evaluation metrics, AHF-U-Net (CLSCA) offers the highest accuracy (96.63%), U-Net-iAFF [15] provides the highest sensitivity (83.40%) and Unext [64] provides the highest specificity. Fig. 8 shows the predicted segmented mask of our proposed methods, as well as a few other methods for breast ultrasound images.

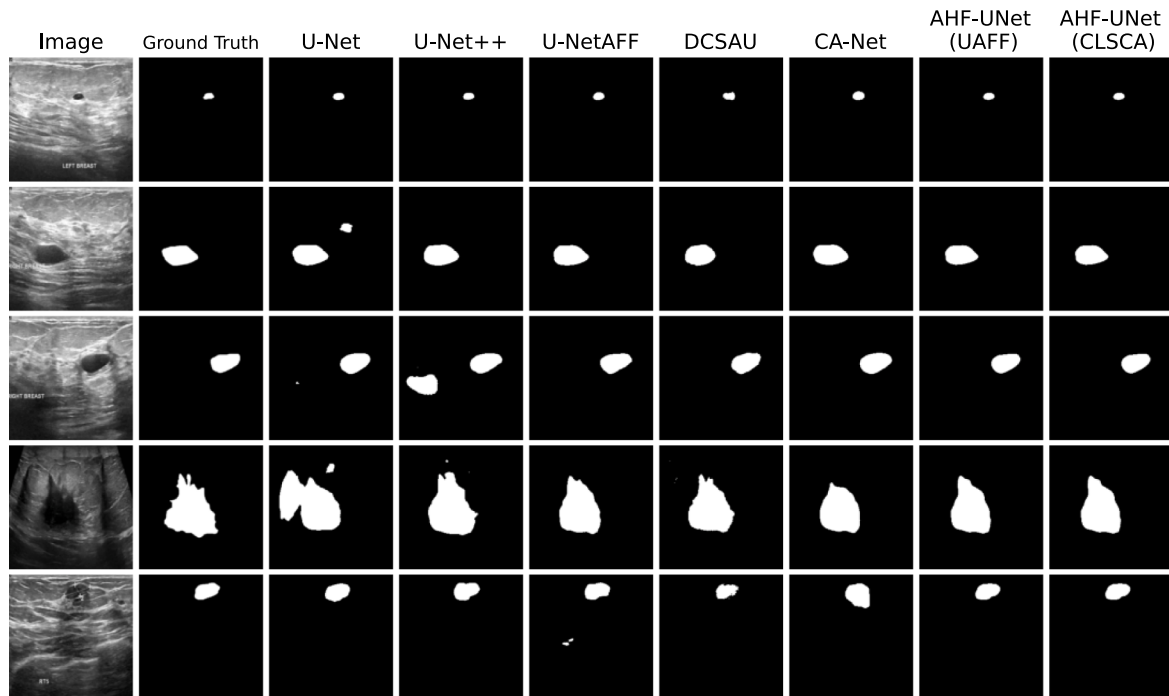
Table 2 shows the quantitative results of the uncertainty-aware versions of the proposed models: UA-AHF-U-Net (UAFF) and UA-AHF-U-Net (CLSCA). To ensure a fair comparison, we needed a model that uses the same uncertainty measurement technique we employed. As Zou et al. [52] already integrated EvidenceCap with U-Net, we also integrated our modified version of EvidenceCap with U-Net and U-Net++. These models were selected due to their popularity in image segmentation, despite their performance not being optimal for our datasets. Additionally, to compare with an attention-based method, we integrated EvidenceCap with U-NetAFF. By doing so, we selected three models from the three major groups into which we divided all the evaluation models. The Table 2 shows that the uncertainty-aware versions perform slightly better than their baselines. However, both UA-AHF-UNet models outperform the other uncertainty-aware models. Specifically, UA-AHF-UNet (CLSCA) leads in IoU (70.33%), Dice score (82.51%), Accuracy (96.89%), and Specificity (98.60%) by 0.21%, 0.08%, 0.26%, and 0.22% respectively, compared to AHF-UNET (CLSCA).

Fig. 9 represents the qualitative results of this experiment showing predicted segmentation with uncertainty maps for each algorithm. We include only one UA-AHF-U-Net's output for qualitative analysis

Table 1

This table represents the quantitative results of our proposed models with other state-of-the-art models for the BUS dataset [22], where **red** denotes the highest score and **green** denotes the second highest. We trained and evaluated all the models in identical settings to ensure a fair and unbiased comparison. To be more specific, image size, batch size, learning rate, number of epochs, and a few other hyperparameters are kept the same for all models.

Methods	IoU	Dice	Accuracy	Sensitivity	Specificity
U-Net [11]	64.61	78.50	96.34	73.86	97.57
SegNet [14]	58.95	65.57	82.13	69.44	93.35
U-Net++ [12]	65.37	79.06	96.07	76.21	97.91
Unetx [64]	67.42	80.59	95.92	80.95	98.80
CUNet [26]	52.78	59.78	89.72	72.52	95.02
CA-Net [65]	66.34	79.77	96.54	78.67	98.64
U-NetAFF [15]	65.03	78.81	95.85	79.29	97.64
U-Net-iAFF [15]	66.32	79.75	95.88	83.40	97.23
TransUnet [66]	66.15	76.71	93.46	75.87	95.72
SwinUnet [67]	68.92	80.53	96.43	81.71	98.11
DCSAU [71]	58.77	74.03	94.88	75.10	97.01
AHF-UNet (UAFF)	69.42	81.95	96.49	82.01	98.05
AHF-UNet (CLSCA)	70.12	82.43	96.63	81.91	98.38

**Fig. 8.** Qualitative results of different models for breast ultrasound images collected from the BUS dataset [22].**Table 2**

Comparison of our proposed uncertainty-aware models with the uncertainty-aware versions of three other models (U-Net, U-Net++, and U-NetAFF) from three major groups for the BUS dataset [22].

Methods	IoU	Dice	Acc	Sens	Spec
UA-UNet	65.33	79.03	96.26	77.91	98.08
UA-U-Net++	65.84	80.60	96.81	77.94	97.91
UA-U-NetAFF	68.84	81.24	95.98	81.83	97.72
UA-AHF-UNet (UAFF)	69.81	82.22	96.53	82.62	98.03
UA-AHF-UNet (CLSCA)	70.33	82.51	96.89	80.34	98.60

because of similar qualitative results between the two UA-AHF-U-Nets. Within the uncertainty map, there are areas represented in dark blue when the ground truth mask matches the predicted mask, and red when the ground truth mask does not match the predicted mask. Hence, the dark blue color indicates low uncertainty or high confidence, while the red color indicates high uncertainty or low confidence. For a clearer explanation, let us analyze the output of U-Net++ for the second image (2nd row) in Fig. 2. A particular area of the ground truth mask

differs significantly from the mask predicted in this image. This area is highlighted in red, indicating that radiologists should double-check it.

5.2. Evaluation on ISIC-16

We also evaluated the performance of AHF-U-Net (UAFF) and AHF-U-Net (CLSCA) for the skin lesion segmentation task hosted by the International Skin Imaging Collaboration (ISIC), 2016. Table 3 presents the quantitative results of our model compared to other state-of-the-art models, and the qualitative results are provided in Fig. 10.

From Table 3, it is noticeable that the methods (methods in rows 3 to 6) that use multiscale skip connections or aggregate multilevel context information are better (except CU-Net) than the methods that use simple skip connections (U-Net and SegNet). In addition, integrating attention mechanisms into image segmentation leads to enhanced segmentation performance. We have implemented three attention-guided approaches: U-Net-AFF, U-Net-iAFF, and DCSAU, and two advanced transformer-based approaches (TransUnet and SwinUnet) and compared their performance on ISIC-2016.

Generally, these methods perform well. Among them, U-Net-iAFF provides the highest IoU (83.75%), DICE (91.15%), and sensitivity

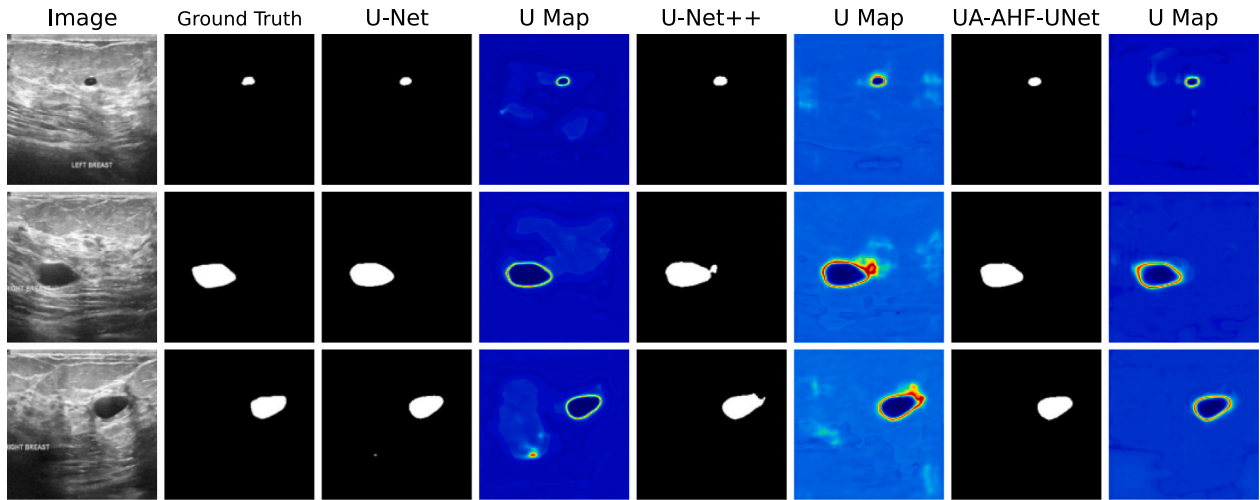


Fig. 9. Predicted segmentation map and corresponding uncertainty map (U Map) generated from uncertainty-aware models for breast ultrasound images collected from the BUS dataset [22].

Table 3

This table represents the quantitative results of our proposed models with other state-of-the-art models for the ISIC-2016 dataset [23], where **red** denotes the highest score and **green** denotes the second highest. We trained and evaluated all the models in identical settings to ensure a fair and unbiased comparison. To be more specific, the image size, batch size, learning rate, number of epochs, and a few other hyperparameters are kept the same for all models.

Methods	IoU	Dice	Accuracy	Sensitivity	Specificity
U-Net [11]	81.27	89.67	94.36	88.82	96.47
SegNet [14]	79.67	85.29	89.66	87.53	90.51
U-Net++ [12]	81.46	89.78	94.74	87.78	97.11
Unetx [64]	81.89	89.91	94.36	87.19	97.02
CUNet [26]	80.67	87.51	93.68	86.01	95.54
CA-Net [65]	82.30	90.29	94.72	89.25	96.68
U-NetAFF [15]	83.01	90.53	95.03	88.88	97.16
U-Net-iAFF [15]	83.75	91.15	95.10	91.72	96.38
TransUnet [66]	79.56	90.15	92.44	82.79	94.91
SwinUnet [67]	82.88	90.03	94.58	89.98	96.71
DCSAU [71]	82.21	90.23	94.74	88.28	97.19
AHF-UNet (UAFF)	83.31	90.89	95.43	91.64	98.31
AHF-UNet (CLSCA)	83.16	90.69	95.06	91.42	96.36

Table 4

Comparison of our proposed uncertainty-aware models with the uncertainty-aware versions of three other models (U-Net, U-Net++, and U-NetAFF) from three major groups for the ISIC-16 dataset [23].

Methods	IoU	Dice	Acc	Sens	Spec
UA-U-Net	81.84	90.01	94.53	89.50	96.44
UA-U-Net++	80.90	89.44	94.61	86.73	97.43
UA-U-NetAFF	82.42	90.36	95.02	88.82	97.23
UA-AHF-UNet (UAFF)	83.68	91.19	95.46	91.69	97.80
UA-AHF-UNet (CLSCA)	83.02	90.76	95.27	91.49	97.97

(91.72%). However, both of our models perform reasonably well; AHF-U-Net (UAFF) achieved the second-highest score in terms of IoU (83.31%), DICE (90.89%), and sensitivity (91.64%) and the highest score in terms of accuracy (95.43%) and specificity (98.31%). We used our proposed attention-aware versions of AHF-U-Net to measure the uncertainty of predicted skin lesion segmentation. Fig. 11 shows the predicted segmentation maps with the corresponding uncertainty maps generated from our model and the uncertainty-aware versions of U-Net and U-Net++. The quantitative results of this experiment are shown in Table 4, where UA-AHF-U-Net (UAFF) outperforms others in all evaluation metrics. Although we benchmarked U-Net-iAFF for ISIC-2016 according to Table 3, our proposed UA-AHF-U-Net (UAFF) provides 0.04%, 0.36%, and 1.42% higher values for DICE (91.19%), accuracy (95.46%), and specificity (97.8%).

5.3. Evaluation on ISIC-17

AHF-U-Net (UAFF) and AHF-U-Net (CLSCA) also performed well on the skin lesion segmentation task hosted by the International Skin Imaging Collaboration (ISIC) in 2017. Table 5 summarizes the quantitative results of our proposed models compared to other mainstream models, while Fig. 12 presents the qualitative results of this experiment.

We first implemented U-Net and SegNet, the most common and widely used image segmentation architectures for evaluation. U-Net (IoU = 81.48% and DICE = 89.79%) performs much better than SegNet (IoU = 79.86% and DICE = 86.78%) in all evaluation metrics. Next, we implemented a few methods that fused multiscale context information: U-Net++, Unetx, CU-Net, and CA-Net. Most of these methods provide approximately 82% IoU and better segmentation results than the previous two methods. The exception occurs for CU-Net, whose performance is relatively low for ISIC-2016 and ISIC-2017. However, attention-guided models and transformer-based models outperform the above techniques for the ISIC-2017 dataset. DCSAU gives the highest IoU (83.49%) and DICE (91.21%) among the eleven standard methods we use for evaluation. Combining the features of attention mechanism and multiscale fusion, both AHF-U-Net (UAFF and CLSCA) versions perform better than DCSAU. Finally, we benchmarked AHF-U-Net (UAFF) for skin lesion segmentation (ISIC-2017) to provide the highest IoU (84.6%), DICE (91.66%) and precision (97.02%).

Table 6 compares the proposed uncertainty-aware version of our model with the uncertainty-aware versions of U-Net, U-Net++, and U-NetAFF. For ISIC-17, our implemented uncertainty-aware versions

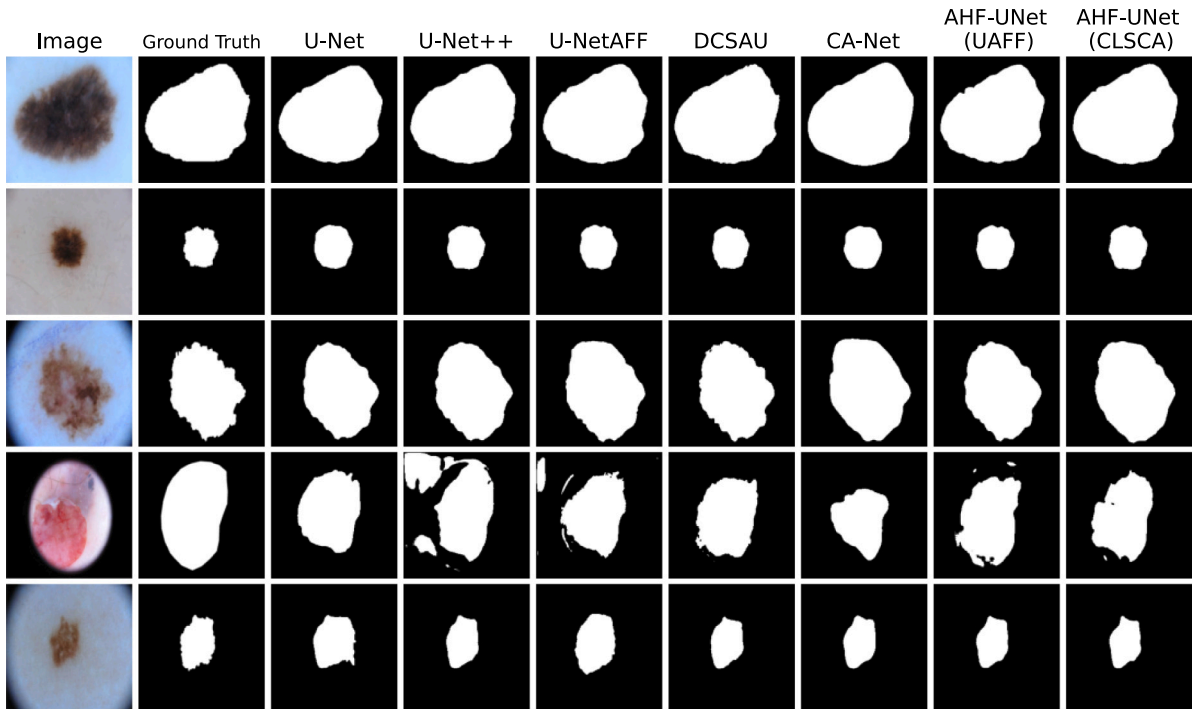


Fig. 10. Qualitative results of different models for skin lesion segmentation collected from the ISIC-2016 dataset [23].

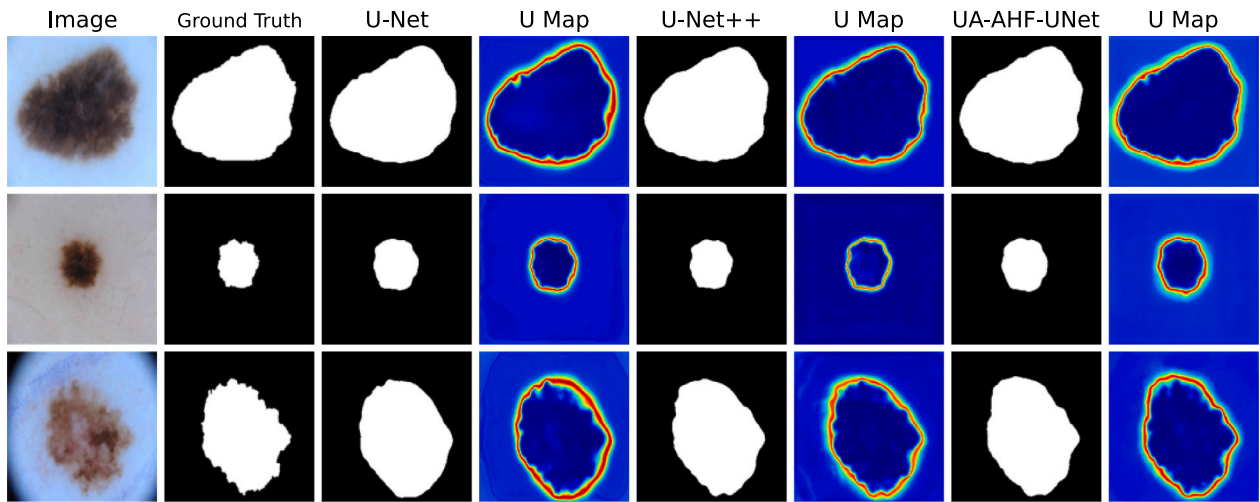


Fig. 11. Predicted segmentation map and corresponding uncertainty map (U-Map) generated from uncertainty-aware models for skin lesion images collected from the ISIC-2016 dataset [23].

of U-Net, U-Net++ and U-NetAFF outperform their original versions. However, two uncertainty-aware versions of our proposed model do not show any improvement compared to their baseline models. In terms of uncertainty-awareness, UA-AHF-U-NET (UAFF) is the best of these five models. Fig. 13 shows the qualitative results of these methods.

5.4. Evaluation of different uncertainty quantification (UQ) methods

We conducted a quantitative analysis of MC Dropout, Deep Ensemble, and EvidenceCap to determine which technique is most effective in our context. Different uncertainty quantification methods can yield varied values to represent uncertainty. To understand these variations, we plotted histograms of the uncertainty values of all the pixels for different methods. This analysis revealed that a single uncertainty threshold does not suffice to compare all methods comprehensively. Consequently, we chose three uncertainty thresholds: 0.4, 0.5, and

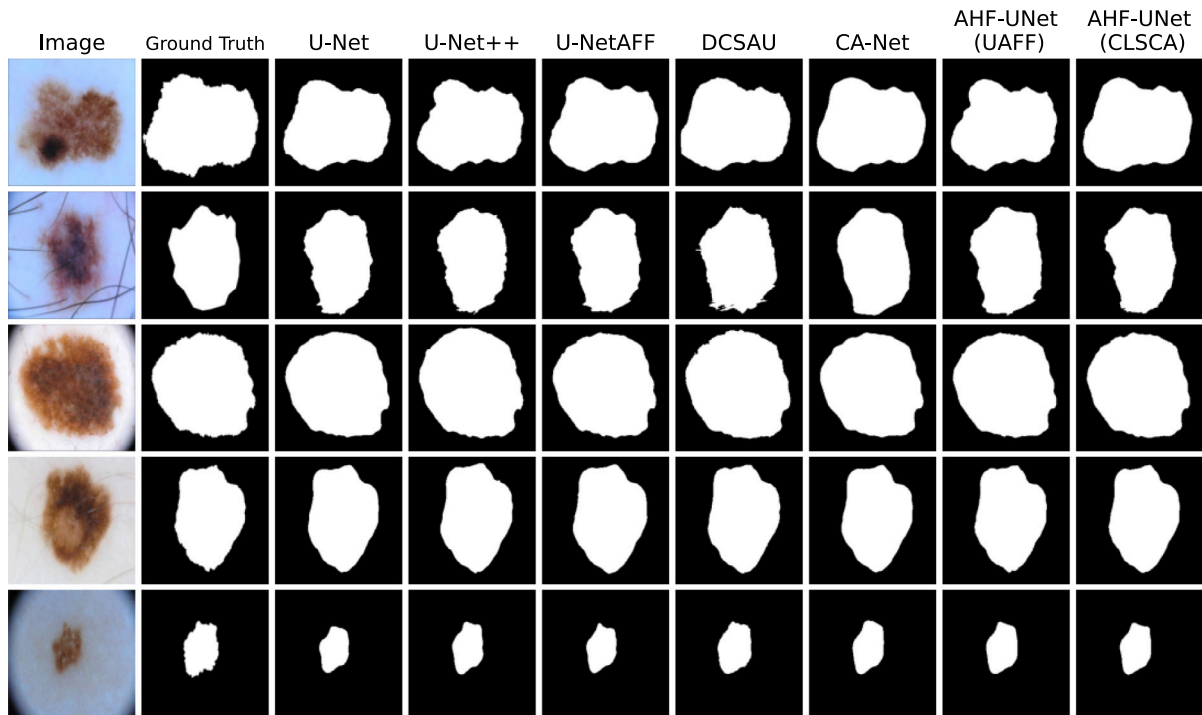
0.6, to implement in our test set. For each threshold, we computed the Intersection over Union (IoU) and Dice scores for all pixels as well as pixels whose uncertainty scores are less than those thresholds only. As expected, separating out certain pixels based on measured uncertainty resulted in higher IoU and Dice scores compared to using all pixels. Table 7 presents the quantitative results for all UQ techniques, contrasting performances for certain pixels versus all pixels within the BUS dataset. Our modified EvidenceCap consistently showed higher IoU and Dice scores for certain pixels for all thresholds compared to the IoU(80%) and Dice(89%) given by all pixels. However, with the threshold of 0.4 yielded the best results (IoU = 96%, Dice = 98%). In contrast, neither MC Dropout nor Deep Ensemble showed improved performance for certain pixels compared to all pixels.

As uncertainty maps provide an intuitive understanding of where the model is most and least certain about its predictions, we analyzed different uncertainty maps generated by various uncertainty

Table 5

This table represents the quantitative results of our proposed models with other state-of-the-art models for the ISIC-2017 dataset [24], where **red** denotes the highest score and **green** denotes the second highest. We trained and evaluated all the models in identical settings to ensure a fair and unbiased comparison. To be more specific, image size, batch size, learning rate, number of epochs, and a few other hyperparameters are kept the same for all models.

Methods	IoU	Dice	Accuracy	Sensitivity	Specificity
U-Net [11]	81.48	89.79	96.23	87.93	98.16
SegNet [14]	79.86	86.78	88.86	86.29	92.01
U-Net++ [12]	82.05	90.38	96.28	88.07	98.15
Unetx [64]	82.93	90.85	95.91	88.95	98.62
CUNet [26]	80.95	88.35	95.12	87.74	96.83
CA-Net [65]	82.70	90.53	96.45	88.41	97.92
U-NetAFF [15]	82.97	88.69	96.61	87.52	98.73
U-Net-iAFF [15]	83.07	90.75	94.82	92.36	95.75
TransUnet [66]	79.42	81.29	92.02	82.91	95.86
SwinUnet [67]	80.76	91.01	96.14	91.36	97.39
DCSAU [71]	83.49	91.21	96.57	87.56	97.95
AHF-UNet (UAFF)	84.6	91.66	97.02	88.46	98.95
AHF-UNet (CLSCA)	84.4	91.54	96.92	89.25	98.68

**Fig. 12.** Qualitative results of different models for skin lesion segmentation collected from the ISIC-2017 dataset [24].**Table 6**

Comparison of our proposed uncertainty-aware models with the uncertainty-aware versions of three other models (U-Net, U-Net++, and U-NetAFF) from three major groups for the ISIC-2017 dataset [24].

Methods	IoU	Dice	Acc	Sens	Spec
UA-UNet	82.52	90.43	96.55	87.98	98.66
UA-U-Net++	82.24	90.44	96.39	88.14	98.98
UA-U-NetAFF	83.04	90.73	96.56	87.98	98.59
UA-AHF-UNet (UAFF)	84.32	91.49	96.93	88.47	98.94
UA-AHF-UNet (CLSCA)	84.01	90.96	96.72	88.19	98.69

quantification techniques for the same inputs [14]. For these maps, we employed the “jet” color map, where blue represents areas of low uncertainty—indicating high model confidence. Conversely, red signifies high uncertainty, marking regions where the model’s confidence is low. Green or yellow colors denote moderate uncertainty, typically appearing in transition zones that highlight areas where the model begins to question its predictions, possibly due to gradients changes in the features being examined.

Fig. 14 presents three uncertainty maps produced by EvidenceCap, MC Dropout, and Deep Ensemble for the BUS dataset. Upon close examination, it is evident that the ground truth and predictions are closely aligned, suggesting the model’s predictions are mostly correct, with only some inconsistency around the borders. Our UQ technique, EvidenceCap, uniquely highlights these areas in green, indicating regions where radiologists should focus more attention. In contrast, the UQ maps from MC Dropout and Deep Ensemble show significant areas of high uncertainty (marked in red) even for correctly classified pixels, which could potentially mislead radiologists. After analyzing both quantitative and qualitative results, we have chosen to implement EvidenceCap for our experiments due to its more accurate and relevant uncertainty visualization.

5.5. Statistical validation of the proposed model

In this research, we extensively evaluated the performance of our proposed segmentation models, AHF-U-Net (UAFF) and AHF-U-Net (CLSCA), across three datasets: BUS, ISIC-16, and ISIC-17. The comparison included 11 other established models. Our models achieved the

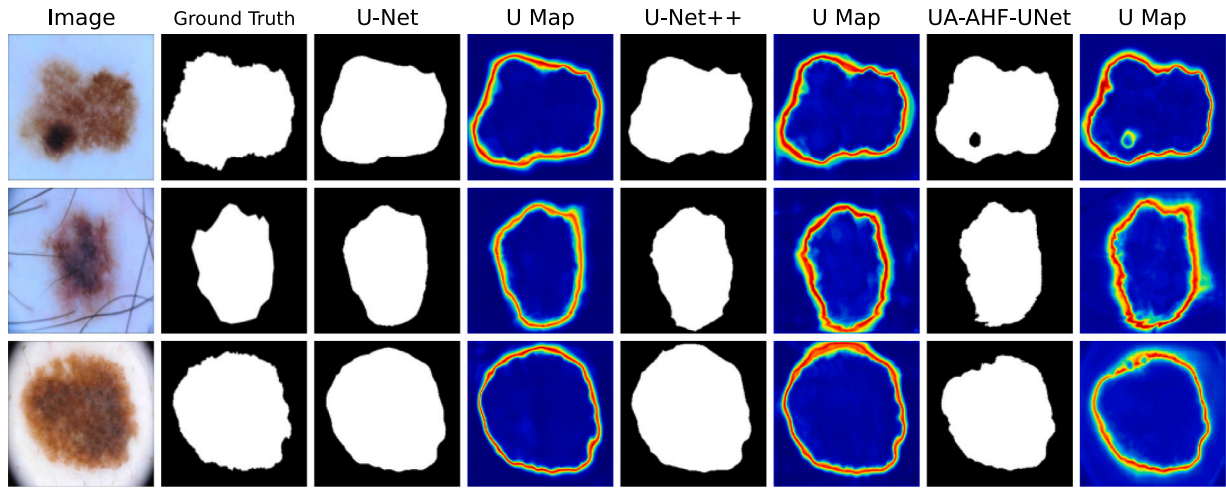


Fig. 13. Predicted segmentation map and corresponding uncertainty map (U-Map) generated by uncertainty-aware models for skin lesion images collected from the ISIC-2017 dataset [24].

Table 7

Quantitative comparison of uncertainty quantification techniques.

Uncertainty Threshold	Pixel Type	EvidenceCap IoU (%)	EvidenceCap Dice (%)	MC Dropout IoU (%)	MC Dropout Dice (%)	Deep Ensemble IoU (%)	Deep Ensemble Dice (%)
N/A	All pixels	80.89	89.44	85.83	92.37	82.33	90.31
0.4	Certain pixels	96.26	98.12	83.69	91.12	80.06	88.92
0.5	Certain pixels	91.15	95.37	85.02	91.90	81.14	89.59
0.6	Certain pixels	86.95	93.02	85.57	92.22	81.60	89.87

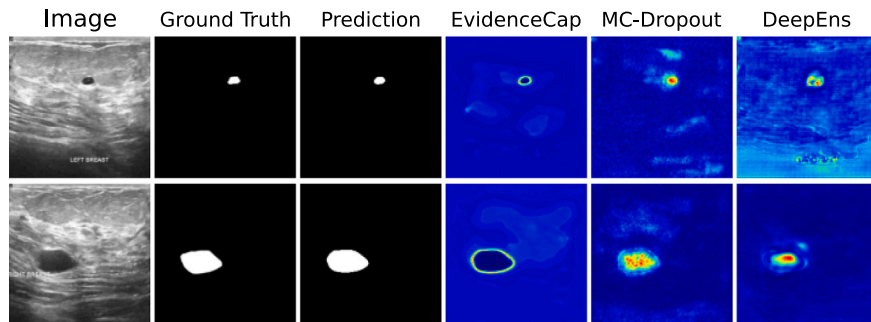


Fig. 14. Comparative uncertainty visualization across different techniques.

highest scores on the BUS and ISIC-17 datasets and the second-highest scores on ISIC-16. To ensure robustness, we ran each model five times. Although we utilized five evaluation criteria for performance evaluation, we used the IoU and Dice for statistical validation, as these are the two most crucial metrics for image segmentation. During evaluation, we benchmarked CLSCA for the BUS dataset and UAFF for the ISIC-17. For ISIC-16, UAFF performed better than CLSCA, though it achieved the second position because U-Net-iAFF achieved the highest score. Consequently, for statistical validation through p -value comparison, we focused on these leading models against the other 11 models for their respective datasets.

To determine which statistical test (paired t-test or Wilcoxon test) is appropriate for calculating the p -value for our data, we first need to assess the normality of our data distributions. Therefore, we first utilized the Shapiro–Wilk test [72] to evaluate the normality of our data. If the p -values from the Shapiro–Wilk test for the results (IoU and Dice) of both models are greater than 0.05, this suggests that the data follow a normal distribution. In such cases, we use the paired t-test to calculate the p -value. Otherwise, we use the Wilcoxon test.

Figs. 15, 16, and 17 are based on the p -values calculated for the BUS, ISIC-16, and ISIC-17 datasets, respectively. If the p -values are less

than 0.05, they are represented with a ✓ symbol; if they are 0.05 or greater, they are represented with a ✗ symbol.

The results are categorized based on the p -values as follows: results with p -values below 0.05 for both IoU and Dice are classified as strongly significant; results where at least one metric has a p -value below 0.05 are considered significant; and results that do not meet this threshold are classified as not significant. This statistical analysis compares our methods with the other compared methods. If we consider the three tables together and count the overall results for statistical significance, we have 21 results that are strongly significant, 3 that are significant, and 9 that are not significant, out of a total of 33 evaluations. Given this distribution, we consider our results to be statistically significant.

5.6. Computational efficiency and effectiveness

We have compared the computational efficiency of both of our proposed algorithms AHF-U-Net (UAFF) and AHF-U-Net (CLSCA), with a range of established methods in medical image segmentation. For each algorithm, we measured the execution time per epoch over a span of 100 epochs. In Fig. 18, the number of epoch varies for different

Methods	IoU	Dice	Decision
U-Net	✓	✓	strongly significant
SegNet	✓	✓	strongly significant
Unet++	✓	✓	strongly significant
Unet	✓	✓	strongly significant
CUNet	✓	✓	strongly significant
CANet	✓	✓	strongly significant
UNetAFF	✓	✓	strongly significant
Unet_iAFF	✓	✓	strongly significant
TransUnet	✓	✓	strongly significant
SwinUnet	✓	✓	strongly significant
DCSAU	✓	✓	strongly significant

Fig. 15. This statistical decision table is based on the p-values of our proposed method versus other methods for the BUS dataset [22]. The ✓ symbol represents p-values less than 0.05, indicating statistical significance, while the ✗ symbol represents p-values greater than or equal to 0.05, indicating a lack of statistical significance.

Methods	IoU	Dice	Decision
U-Net	✓	✓	strongly significant
SegNet	✓	✓	strongly significant
Unet++	✓	✓	strongly significant
Unet	✗	✗	not significant
CUNet	✓	✓	strongly significant
CANet	✗	✗	not significant
UNetAFF	✗	✗	not significant
Unet_iAFF	✗	✗	not significant
TransUnet	✓	✗	significant
SwinUnet	✗	✗	not significant
DCSAU	✗	✗	not significant

Fig. 16. This statistical decision table is based on the p-values of our proposed method versus other methods for the ISIC-16 dataset [23]. The ✓ symbol represents p-values less than 0.05, indicating statistical significance, while the ✗ symbol represents p-values greater than or equal to 0.05, indicating a lack of statistical significance.

Methods	IoU	Dice	Decision
U-Net	✓	✓	strongly significant
SegNet	✓	✓	strongly significant
Unet++	✓	✓	strongly significant
Unet	✗	✗	not significant
CUNet	✓	✓	strongly significant
CANet	✓	✓	strongly significant
UNetAFF	✗	✓	significant
Unet_iAFF	✗	✗	not significant
TransUnet	✓	✓	strongly significant
SwinUnet	✓	✗	significant
DCSAU	✗	✗	not significant

Fig. 17. This statistical decision table is based on the p-values of our proposed method versus other methods for the ISIC-17 dataset [24]. The ✓ symbol represents p-values less than 0.05, indicating statistical significance, while the ✗ symbol represents p-values greater than or equal to 0.05, indicating a lack of statistical significance.

algorithms because we applied an early stopping mechanism, which terminates training if no improvement is observed in the IoU score of the segmentation result for 20 consecutive epochs.

As a result of our analysis, it is observed that U-Net++ requires the highest execution time among the tested algorithms. In contrast, Unet showed the lowest computational demand, making it the most efficient algorithm in terms of execution time. Although in the field of medical image analysis, precise image segmentation is prioritized over computational efficiency because precision in segmentation significantly impacts clinical outcomes. In most cases, delays in processing

times are acceptable if they result in more accurate and clinically useful information.

Our algorithms, AHF-U-Net (UAFF) and AHF-U-Net (CLSCA), while not achieving the lowest execution times, demonstrated precise image segmentation. Both algorithms did not reach the maximum number of epochs due to the early stopping mechanism. This indicates a well-balanced approach to model training that conserves computational resources without compromising effectiveness. This early stopping mechanism not only prevents overfitting but also optimizes the

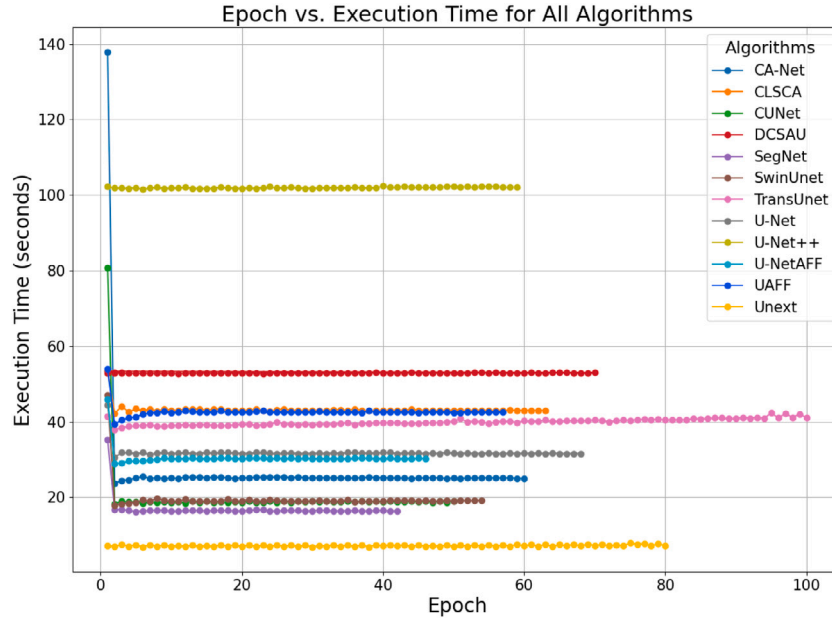


Fig. 18. Comparative analysis of execution times across epochs for various segmentation algorithms using the BUS dataset [22].

training duration by terminating the process when there is no further improvement in model performance.

5.7. Explainability overview

To ensure that our proposed model is properly focusing or not on the relevant pixels corresponding to the target object, we utilized Grad-CAM [73]. It is a powerful visual tool representing transparency and interpretability in model predictions. It highlights where the model directs its attention for predictions, and viewing the heatmap enables us to understand whether the model is focusing on the appropriate features. Such visualization is crucial in applications such as medical imaging, which helps build radiologists' trust and prevent misdiagnosis. We applied Grad-CAM to all datasets that we used in this research. Fig. 19 shows the heat map generated after applying Grad-CAM to the layer immediately below the final layer of AHF-U-Net (UAFF). We use the 'jet' colormap to visualize the heatmaps, where red hues indicate the highest values or the most significant regions of the image, while blue indicates the opposite. For example, the first two rows of Fig. 1 represent the image (collected from BUS [22]) and its corresponding heat maps for BUS datasets. On the heat map, red corresponds to the most significant region or tumor presence, and blue corresponds to the less critical region or background. The tumor position on the image aligns precisely with the red heightened areas of interest on the heatmap, as shown in Fig. 19. Furthermore, the highlighted objects show minimal background distractions. These effects result from adding global spatial attention, global channel attention, and local channel context. In summary, by applying Grad-CAM, we can understand the reasoning behind the model's predictions, strengthening our models' interpretability and trustworthiness. This also mitigates the gap between efficiency and explainability in AI-based medical imaging.

5.8. Ablation study

The proposed AFF-U-Net is built by integrating EAF and DAF modules with U-Net. To evaluate the contribution of each module, we excluded them from the network one by one and analyzed the outputs. Table 8 summarizes the results of two main evaluation metrics (IoU and DICE) for three datasets in three different network configurations.

As AFF-U-Net has one EAF module and two types of DAF modules (UAFF and CLSCA), we have designed the following configurations for ablation experiments:

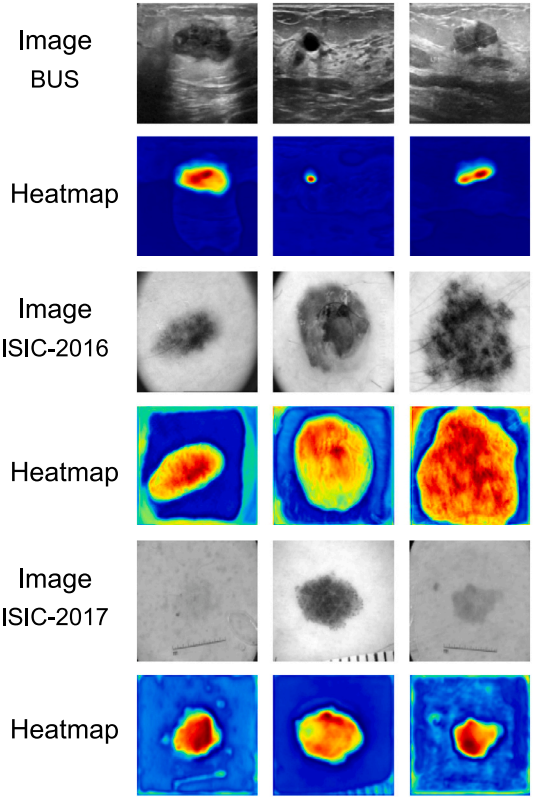


Fig. 19. First row shows the images collected from BUS [22] dataset, and the second row illustrates the corresponding heatmaps generated after applying Grad-CAM to the second-to-last layer of our architecture. The third and fourth rows represent the image and heatmap for the ISIC-16 [23] dataset, while the last two represent the ISIC-17 dataset [24].

- **Conf1** — deactivate the EAF module while using UAFF as the DAF module.

Table 8

Comparison of IoU and DICE for ablation experiments with different AHF-U-NET configurations, where blue values represent the highest score.

Methods	BUS		ISIC-16		ISIC-17	
	IoU	Dice	IoU	Dice	IoU	Dice
Conf1: Deactivate the EAF module (UAFF)	68.35	81.13	82.78	90.06	83.09	90.67
Conf2: Deactivate the EAF module (CLSCA)	68.41	81.5	81.56	89.24	83.71	90.25
Conf3: Deactivate the DAF module	68.57	81.35	82.96	90.17	83.80	90.82
AHF-U-NET (UAFF)	69.42	81.95	83.31	90.89	84.60	91.66
AHF-U-NET (CLSCA)	69.84	82.24	83.16	90.69	84.40	91.54

- **Conf2** — deactivate the EAF module while using CLSCA as the DAF module.
- **Conf3** — deactivate the DAF module.

Another key feature of our model is the Hierarchical Attention-Enhanced (HAE) skip connection generated by the EAF module. Therefore, the HAE is automatically removed when we deactivate the EAF module. Among these three configurations, Conf3 outperforms the other two, indicating the impact of the HAE skip connection. Furthermore, the IoU and DICE of Conf1 and Conf2 indicate a significant decrease in segmentation performance. The entire network has approximately 1% higher performance than these results, which is true for all datasets. For the BUS dataset, AHF-U-NET (CLSCA) provides the highest IoU (69.84%) and DICE (82.24%) while Conf1 offers the lowest IoU (68.35%) and DICE (81.13%). We benchmarked AHF-U-Net(UAFF) for ISIC-2016 and ISIC-2017. The highest IoU and DICE for ISIC-2016 are 83.11% and 90.78%, respectively. The lowest values are provided by Conf2 (IoU = (81.56%) and DICE = (89.24%)). For ISIC-2017, the highest IoU and DICE are 84.46% and 91.36%, respectively, while Conf1 gives the lowest IoU (83.40%) and Conf2 gives the lowest DICE (90.25%).

6. Conclusion and future work

Our proposed AHF-U-Net incorporates multilevel attention-guided feature fusion on both the encoder and decoder sides of the U-Net architecture. Additionally, we integrate simple and hierarchical skip connections to transfer precise spatial information from the encoder to the decoder branches. By combining these two mechanisms, our network can focus on both “where (spatial)” and “what (channel)” is relevant in the image, which is a key requirement for segmenting intricate medical images. Furthermore, we also present an uncertainty-aware version of the proposed model, UA-AHF-U-Net. Along with segmentation, this model provides the segmentation uncertainty. This is crucial in medical imaging to guide radiologists in decision-making. We conducted a detailed ablation study to emphasize the impact of the EAF and DAF modules on the proposed model. Although the performance difference between the proposed model and the EAF or DAF exclusion is close to 1%, such a margin is not negligible in the context of medical image analysis. To provide the explainability of the predicted result, we utilize Grad-CAM which provides meaningful visual evidence to ensure that our proposed model appropriately focuses on the relevant regions of interest. For experimental analysis, we have taken three medical image datasets (i.e., Breast Ultrasound, ISIC16, ISIC17) and evaluated our model with eleven previous state-of-the-art models. Based on the two main evaluation metrics (IoU and DICE) selected for this investigation, our model scored the highest for the breast ultrasound dataset, the ISIC17 dataset, and the second highest for the ISIC16 dataset. To be more specific we benchmarked AHF-U-Net (CLSCA) for the breast ultrasound dataset and achieved 70.12% IoU and 82.43% DICE; for ISIC-16 we benchmarked U-Net-IAFF which achieved the highest IoU (83.75%) and DICE (91.15%), while the second best performance is given by AHF-U-Net (UAFF) at 0.44% and 0.26% lower, respectively; for ISIC-17, AHF-U-Net (UAFF) performs best with 84.6% IoU and 91.66% DICE. None of the eleven benchmarking models works with the predicted results’ uncertainty estimate. Hence, to compare the performance of

AHF-U-Net, we implement the uncertainty-aware versions of U-Net, U-Net++, and U-NetAIF. Experimental results indicate that UA-AHF-U-Net outperforms UA-U-Net UA-U-Net++ and UA-U-NetAIF for all datasets.

Although we used spatial attention, channel attention, and attention-enhanced skip connections in this research, we did not include the transformer approach. Beyond their success in Natural Language Processing (NLP), transformers are now gaining significant impact in image segmentation and classification. Integrating transformers into our model could enhance the segmentation process by emphasizing the ability to handle complex dependencies on image data. The idea can be considered for future research. Furthermore, this study does not consider domain adaptation techniques. To ensure the generalizability of segmentation models to recognize the diversity of medical imaging contexts (MRI, CT, and ultrasound images), significant focus can be placed on domain adaptation techniques. Also, to provide a more comprehensive evaluation of the model, expanding the dataset with more diverse and challenging cases would be beneficial. Additionally, in this research, we have analyzed a few uncertainty quantification techniques and found EvidenceCap suitable for our case, but it might not be an optimal solution for all medical imaging challenges. Due to the highly variable nature of medical images and the presence of rare features, investigating appropriate uncertainty quantification techniques for specified tasks could be a salient area of future research.

In conclusion, while our proposed methodologies demonstrate strong performance in 2D medical image segmentation, they are not without limitations. One of the main constraints is that our models have not yet been applied to 3D datasets, which are increasingly common in clinical practice. Therefore, extending our models to 3D datasets will be a focus of our future research. Another technical limitation is our model’s memory efficiency. The integration of attention mechanisms in both the encoder and decoder components, while beneficial for segmentation accuracy, makes our models computationally heavy. As a result, they may not be the most memory-efficient solutions, particularly for deployment in resource-constrained environments. Future work will also explore optimizing these architectures to enhance their efficiency without compromising performance.

CRedit authorship contribution statement

Afsana Ahmed Munia: Writing – original draft, Visualization, Resources, Methodology, Investigation, Formal analysis, Data curation. **Moloud Abdar:** Writing – review & editing, Validation, Supervision, Methodology, Investigation, Conceptualization. **Mehedi Hasan:** Validation, Resources, Methodology, Investigation, Formal analysis. **Mohammad S. Jalali:** Writing – review & editing, Validation, Supervision. **Biplab Banerjee:** Writing – review & editing, Supervision. **Abbas Khosravi:** Writing – review & editing, Validation, Supervision. **Ibrahim Hossain:** Writing – review & editing, Supervision. **Huazhu Fu:** Writing – review & editing, Supervision. **Alejandro F. Frangi:** Writing – review & editing, Validation, Supervision.

Declaration of competing interest

We declare that there are no conflicts of interest to report.

Data availability

Data will be made available on request.

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